

BINOJ MADHURANGA

Software Engineer Intern

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PROFILE

Results driven Software Engineer with hands on experience designing and developing production-grade digital solutions across the full stack. Proficient in Java with Spring Boot, C# with ASP.NET Core, Go (Golang), and Node.js with Express.js for building scalable backend systems and RESTful APIs. Skilled in frontend and mobile application development using React, Next.js, React Native, and Expo, with strong knowledge of SQL based database systems. Possesses a solid understanding of software design patterns and modern software engineering practices, with a focus on building reliable, maintainable, and scalable applications.

EDUCATION

BSc. Honours Degree in Software Engineering Sabaragamuwa University of Sri Lanka	2023 - Present (Expected Year - 2027) Current GPA: 3.24 / 4.00
G.C.E Advanced Level - Physical Science Stream Bandaranayake Central College - Veyangoda	2019 - 2021 Z-Score - 1.332

EXPERIENCE

Advanced Full-Stack Development Program (Training Program)

- Completed the Advanced Full-Stack Development Diploma at ACPT, gaining practical experience in modern backend and database technologies. Developed strong skills in Java, Spring Boot, Hibernate/JPA, and MySQL along with hands on knowledge of API development, authentication, and deployment. This program enhanced my ability to build scalable, secure, and production-ready full-stack applications.

Industry Oriented Software Project Development

- Delivered two production-grade software platforms. SmartBursary, a live bursary management system now actively used by Sabaragamuwa University of Sri Lanka, and Gemora, a commercially-viable AI-powered gemstone trading marketplace. Both products were built end-to-end under joint academic and industry mentorship, meeting real client requirements and shipping to actual users.

PROJECTS

Gemora - AI Powered Digital Gemstone Trading & Auction Platform (Research Project)



- Technologies used: Spring Boot, Tensorflow, React, React Native(Expo), MySQL, Zustand
- A secure digital marketplace for gemstone trading featuring real time auctions, identity verification, socket base messaging and AI-powered analysis. Built frontend with React + React Native and backend with Spring Boot and MySQL. Designed complete UI/UX in Figma.
- A research paper based on this project was published and presented at a conference in Colombo. 

AgriTrust - Blockchain Based Agricultural Verification Platform (Individual)



- Technologies used: ASP.NET Core Web API, SQL Server, Entity Framework Core, Solidity, Next.js, Tailwind CSS, Redux Toolkit, Blockchain Technology
- AgriTrust is a secure agricultural trust platform that enables farmer registration, digital certificate issuance, and transparent verification of farmer authenticity.
- Implemented role based approval workflows, certificate lifecycle management, blockchain hash verification, and scalable REST APIs for secure data integrity.

SmartBursary - Student Bursary Management System (Group Leader)



- Technologies used: Express.js, Next.js, Tailwind CSS, Redux Toolkit, MySQL, Sequelize ORM
- Engineered and shipped a production grade bursary management platform for Sabaragamuwa University of Sri Lanka, replacing manual workflows with a fully digital system handling application submission, multi-level verification, approval routing, and payment tracking at institutional scale.
- Deployed to university infrastructure on Node compatible servers behind a secure REST API layer, with role-based access control, maintainable modular architecture, and CI ready codebase built for long term institutional ownership.

SmartBiz - Open AI integrated ERP System (Individual)



- Technologies used: Spring Boot, MySQL, Hibernate, React, Tailwind CSS, OpenAI API
- Built a full-stack ERP solution for small businesses, covering end-to-end sales management, customer relationship tracking, and inventory control with OpenAI-powered report generation and automated content creation baked into the core workflow.
- Architected with a modular, production-ready structure featuring JWT authentication, role-based access control, transactional email integration, and clean REST endpoints designed to handle real business operations reliably.

FinanceMind - AI powered Finance Tracking Application (Individual)



- Technologies used: Express.js, Flask, PostgreSQL, Sequelize, Next.js + Redux, AWS EC2, NGINX
- AI-powered personal finance tracking system allows users to efficiently manage income, expenses, categories, and monthly financial summaries through a clean and responsive dashboard.
- Provisioned backend microservices on AWS EC2 behind NGINX reverse proxy with an automated CI/CD pipeline; served the frontend via Vercel for global edge distribution.

Email-Ms (University Project Management System)



- Technologies used: Node.js, Express.js, MongoDB, React, Tailwind CSS
- Developed a university-commissioned project management system for academic staff, digitizing project tracking, team coordination, and communication workflows across the institution's academic administration.
- Deployed the Node.js backend on AWS EC2 behind an NGINX reverse proxy, delivering a stable, production-grade hosting environment scoped to the university's performance and operational standards.

CERTIFICATIONS

- **AI Engineer Path** – Scrimba
- **AWS Developer** – Udemy
- **Blockchain Basics** – NASBA
- **Golang** – KodeKloud
- **Intro to Machine Learning** – Kaggle
- **Jenkins Project** – KodeKloud
- **Next.js Crash Course** – Udemy
- **n8n: Zero to Hero** – KodeKloud
- **Postman API Foundations** – Postman
- **SOLID Principles in System Design** – LinkedIn Learning
- **C# Bootcamp** - Udemy

TECHNICAL SKILLS

- **Programming Languages** : Java, C#, JavaScript, TypeScript, SQL, Python , C, PHP, Golang
- **Frameworks & Libraries** : SpringBoot, .ASP.NET Core, Node.js, Express.js, React.js, React Native (Expo), Next.js, Redux, Zustand
- **Databases** : MySQL, PostgreSQL, MongoDB, Sequelize ORM, Hibernate
- **Cloud & Deployment** : AWS EC2, AWS S3, GitHub Actions , Docker, Kubernetes
- **Concepts & Practices** : REST API , MVC architecture, SOLID Principles, JWT Authentication, OOP, Agile/Scrum, CI/CD
- **AI & Automation** : Generative AI integration, NumPy, Pandas, TensorFlow, Machine Learning, CNN, Image Classification, Data Preprocessing

SOFT SKILLS

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|-------------------|-------------------------|-----------------|-------------------|
| • Teamworking | • Leadership | • Documentation | • Communication |
| • Problem-Solving | • Self-Learning Ability | • Presentation | • Time Management |

AWARDS & ACHIEVEMENTS

- Participated in Innovate with Ballerina Coding Challenge conducted by IEEE Student Branch of University of Moratuwa in collaboration with WSO2, demonstrating problem solving and coding skills using Ballerina (2025). 
- Selected as a Semi-Finalist in SLIoT Challenge 2026, a national level IoT innovation competition organized by IESL Student Branch – University of Moratuwa in collaboration with SLT MOBITELE.
- Published research paper “DeepMineAI: A Deep Learning-Based Gemstone Classification Model for Digital Gem Trade Platforms” at The Colombo Conference 2026, presenting an AI driven gemstone classification approach using deep learning. 

VOLUNTEERING

- Career Compass - IEEE Student Branch, Sabaragamuwa University of Sri Lanka -_Public Speaker 
- HOPE - IEEE WIE Student Branch, Sabaragamuwa University of Sri Lanka - Instructor 
- COMURS 2025 - Sabaragamuwa University of Sri Lanka - Programme Moderator 
- PearlHack 4.0 - IEEE Student Branch, Sabaragamuwa University of Sri Lanka - Finance team
- IEEE Day - IEEE Student Branch, Sabaragamuwa University of Sri Lanka - Programme Moderator 

REFERENCES

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